LESSON 6.12

Least Common Multiple

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the least common multiple (LCM) of two numbers by listing or comparing their multiples.

WHY THIS MATTERS Astronaut Reyes checks the oxygen system every 4 hours. Astronaut Chen checks the water recycler every 6 hours. They both start their shifts at 6:00 AM. When is the next time both astronauts will be checking their systems at the same time?

REMEMBER

Least Common Multiple

Least Common Multiple (LCM) A multiple is what you get when you count by a number: 4, 8, 12, 16. The least common multiple is the smallest multiple that two numbers share.

1. Skip count multiples of each number starting with the larger number.
2. Identify the first non-zero multiple that appears in both lists.
3. Prime factorization method: Product of highest powers of all prime factors

Words: **Least Common Multiple** ***(Mínimo común múltiplo)*** **Multiple** ***(Múltiplo)*** **Common multiple** ***(Múltiplo común)*** **Skip counting** ***(Conteo salteado)*** **Prime factorization** ***(Factorización prima)***

1 **Circle the letter of the best answer. Show how you know.**

A board 6 feet long is cut into pieces 3/4 foot long. How many pieces are there?

(A) 2 pieces
(B) 4 1/2 pieces
(C) 8 pieces
(D) 24 pieces

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

A recipe needs 2/3 cup of oats per serving. How many servings can be made from 4 cups?

(A) 1 1/2 servings
(B) 2 2/3 servings
(C) 6 servings
(D) 8 servings

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A runner covers 2 1/2 miles in 1/2 hour. What is the speed in miles per hour?

(A) 1 1/4 miles per hour
(B) 2 miles per hour
(C) 3 miles per hour
(D) 5 miles per hour

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which equation solves 'How many 1/8-cup scoops are in 3/4 cup?'

(A) 1/8 ÷ 3/4
(B) 3/4 ÷ 1/8
(C) 3/4 × 1/8
(D) 3/4 + 1/8

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Hot dogs come in packs of 8 and buns come in packs of 6. You want to buy the exact same number of hot dogs and buns with none left over. What is the LCM of 8 and 6?

(A) 12
(B) 14
(C) 24
(D) 48

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Hot dogs come in packs of 8 and buns come in packs of 6. You want to buy the exact same number of hot dogs and buns with none left over. How many packs of hot dogs do you buy?

(A) 2
(B) 3
(C) 4
(D) 6

SHOW YOUR WORK

EXPLAIN YOUR THINKING

What is the least common multiple of 4 and 6? How did sorting the multiples help you find it?

SENTENCE STARTER The LCM of 4 and 6 is ___ because it is the smallest number in the ___ group.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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WHY THIS MATTERS Astronaut Reyes checks the oxygen system every 4 hours. Astronaut Chen checks the water recycler every 6 hours. They both start their shifts at 6:00 AM. When is the next time both astronauts will be checking their systems at the same time?

PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Stop 3 — Engineering Deck: Match each pair of station systems to the LCM of their cycle times (when they next sync).

_____ Solar panels (LCM 6 & 10 min)	(A) 30
_____ Coolant pumps (LCM 5 & 8 min)	(B) 40
_____ Comms relay (LCM 3 & 9 min)	(C) 9
_____ Lab fans (LCM 4 & 6 min)	(D) 12

2 **Write the name of the correct group on each line.**

Stop 4 — Mission Control: Each clue points to your next stop. Sort each crew schedule by whether the two systems sync again in UNDER 25 minutes or 25 MINUTES OR MORE. Skip count to find each LCM.

Groups: **Sync again in under 25 minutes** | **Sync again in 25 minutes or more**

_____ Reactor checks: every 4 and 6 min

_____ Window shutters: every 8 and 12 min

_____ Galley ovens: every 2 and 7 min

_____ Solar arrays: every 6 and 10 min

_____ Coolant pumps: every 5 and 8 min

_____ Antenna sweeps: every 4 and 9 min

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Hot dogs come in packs of 8 and buns come in packs of 6. You want to buy the exact same number of hot dogs and buns with none left over. How many packs of buns do you buy?

(A) 3
(B) 4
(C) 5
(D) 6

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

On the space station, the air filter runs every 4 min and the water pump runs every 6 min. They just ran together. In how many minutes do they run together again?

(A) 6
(B) 10
(C) 12
(D) 24

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

One supply pod docks at the space station every 8 min, and another docks every 12 min. They just docked together. In how many minutes do they dock together again?

(A) 16
(B) 20
(C) 24
(D) 48

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Level 2 Extension — Crew Member's Schedule Sync Mistake

1. **Problem** The water recycler cycles every 8 minutes and the heat exchanger cycles every 12 minutes. They just ran together. When do they next sync?
2. **List multiples of 8** 8, 16, 24, 32
3. **List multiples of 12** 12, 24, 36
4. **Pick the number to use** The greatest factor that 8 and 12 share is 4.
5. **Answer** They next sync in 4 minutes.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What is the least common multiple of 4 and 6? How did sorting the multiples help you find it?

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